Ensemble Methods

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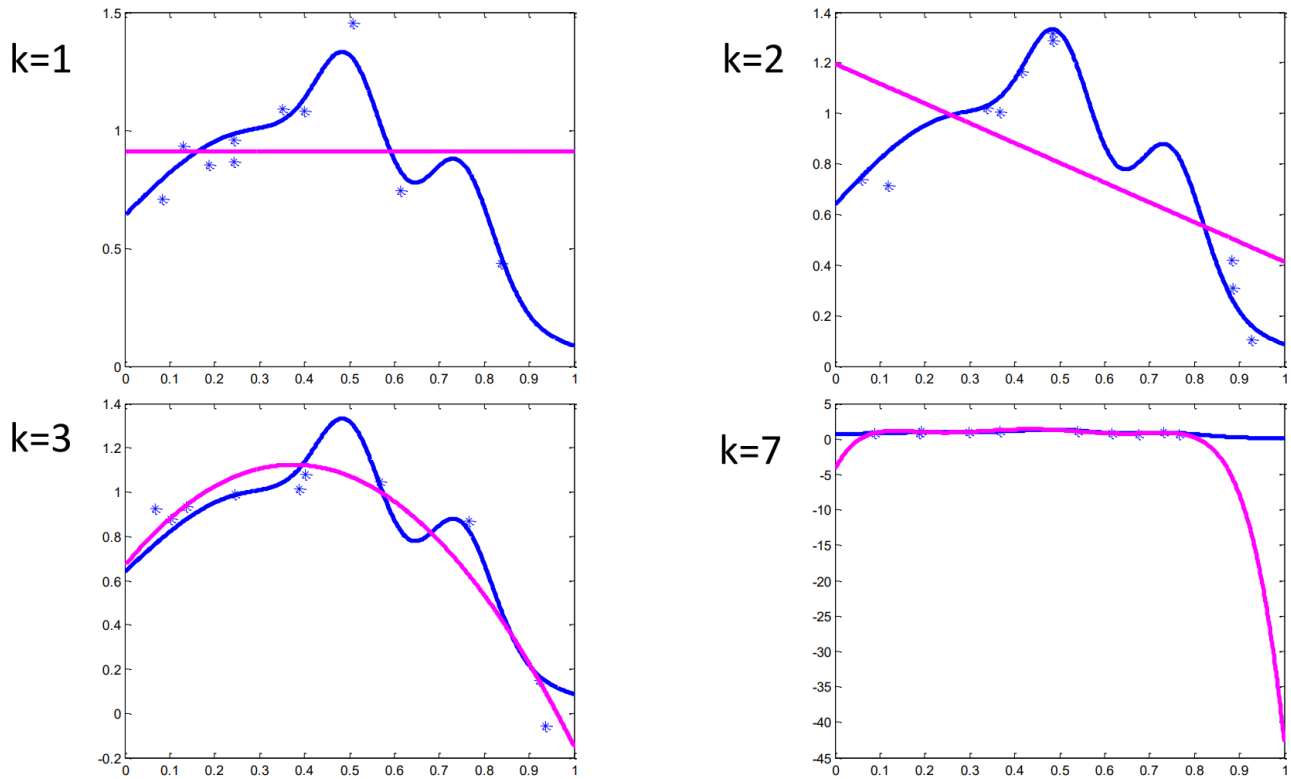
High Variance or Overfitting

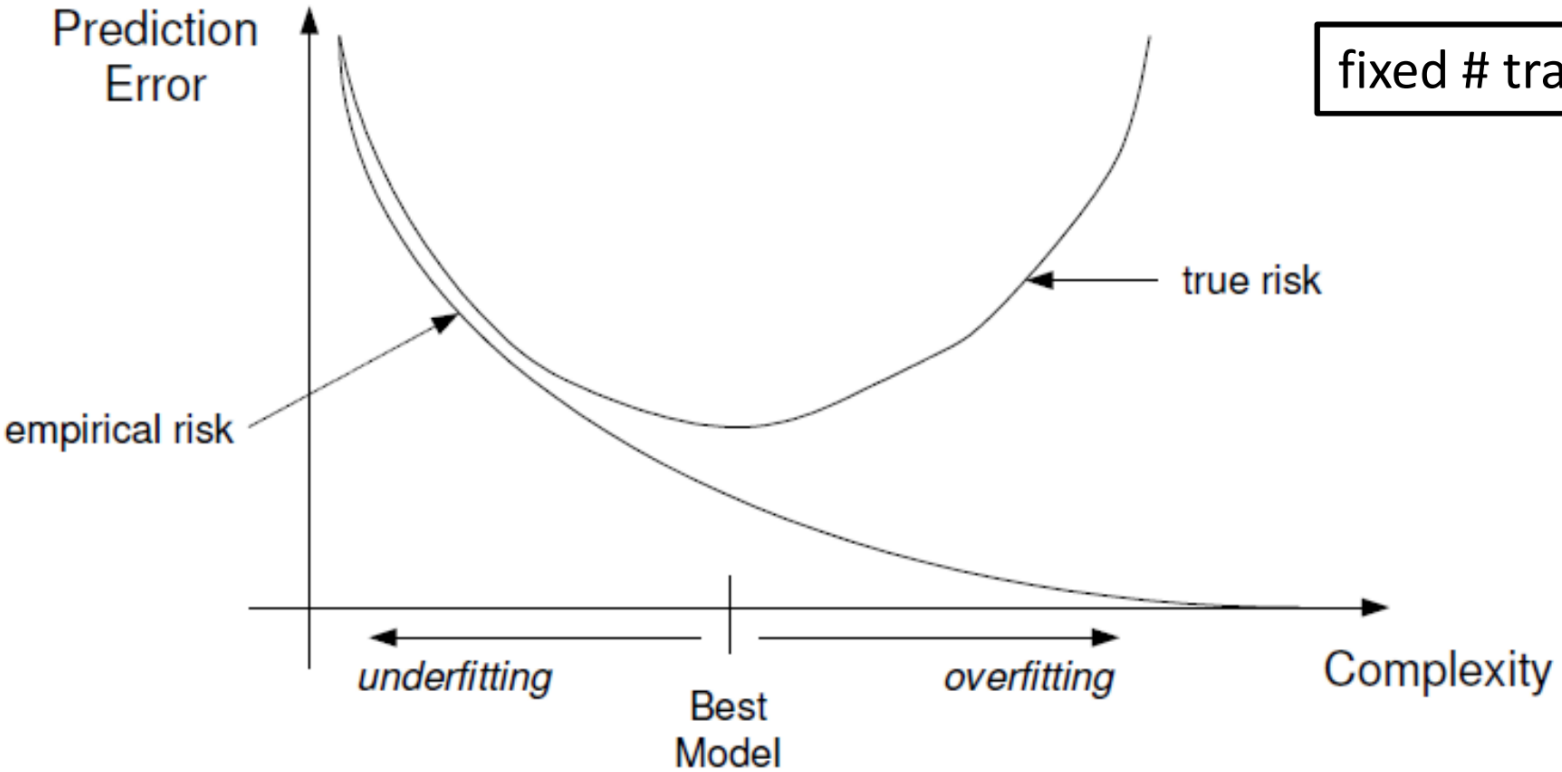
If we allow very complicated predictors, we could overfit the training data.

Examples: Regression (Polynomial of order k – degree up to k-1)



Effect of Model Complexity

If we allow very complicated predictors, we could overfit the training data.



Bias, Variance, and Noise

This gives us the decomposition of expected test error as follows

$$\underbrace{E_{\mathbf{x},y,D}\left[\left(h_D(\mathbf{x})-y\right)^2\right]}_{\text{Expected Test Error}} = \underbrace{E_{\mathbf{x},D}\left[\left(h_D(\mathbf{x})-\bar{h}(\mathbf{x})\right)^2\right]}_{\text{Variance}} + \underbrace{E_{\mathbf{x},y}\left[\left(\bar{y}(\mathbf{x})-y\right)^2\right]}_{\text{Noise}}$$

$$+ \underbrace{E_{\mathbf{x}}\left[\left(\bar{h}(\mathbf{x})-\bar{y}(\mathbf{x})\right)^2\right]}_{\text{Bias}^2}$$

Variance: Captures how much your classifier changes if you train on a different training set. How "over-specialized" is your classifier to a particular training set (overfitting)? If we have the best possible model for our training data, how far off are we from the average classifier?

Bias: What is the inherent error that you obtain from your classifier even with infinite training data? This is due to your classifier being "biased" to a particular kind of solution (e.g. linear classifier). In other words, bias is inherent to your model.

Noise: How big is the data-intrinsic noise? This error measures ambiguity due to your data distribution and feature representation. You can never beat this, it is an aspect of the data.

Bias, Variance, and Noise

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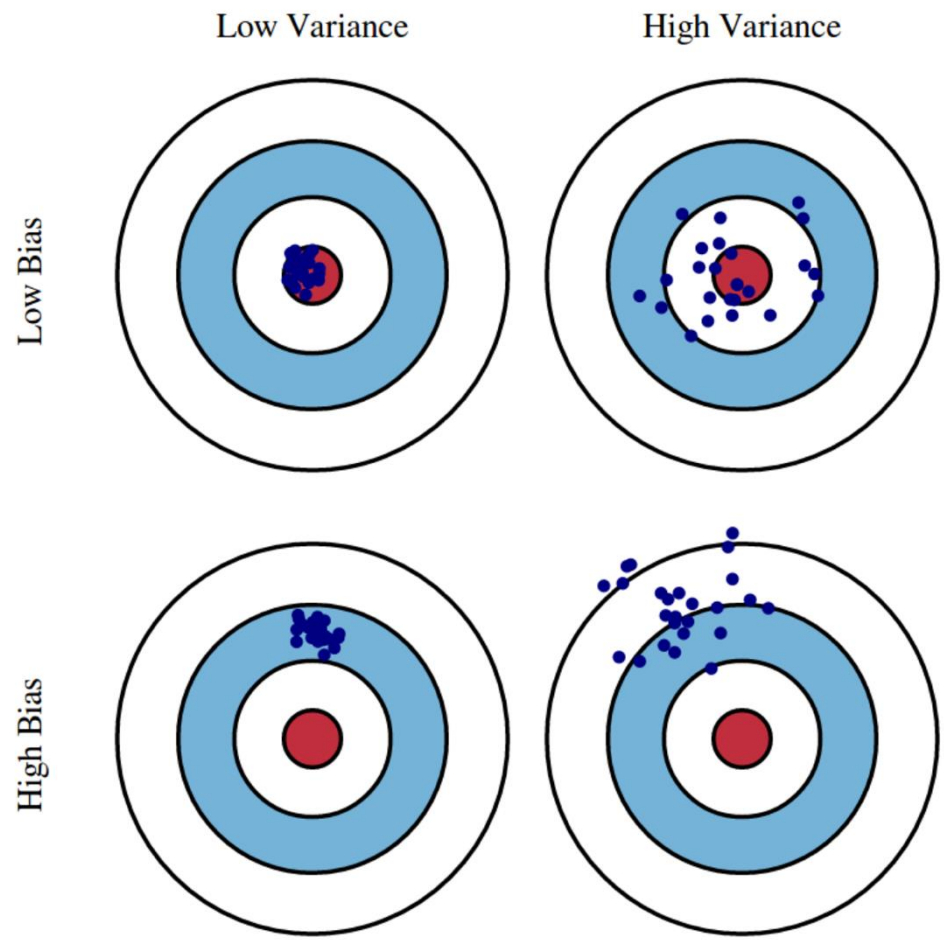
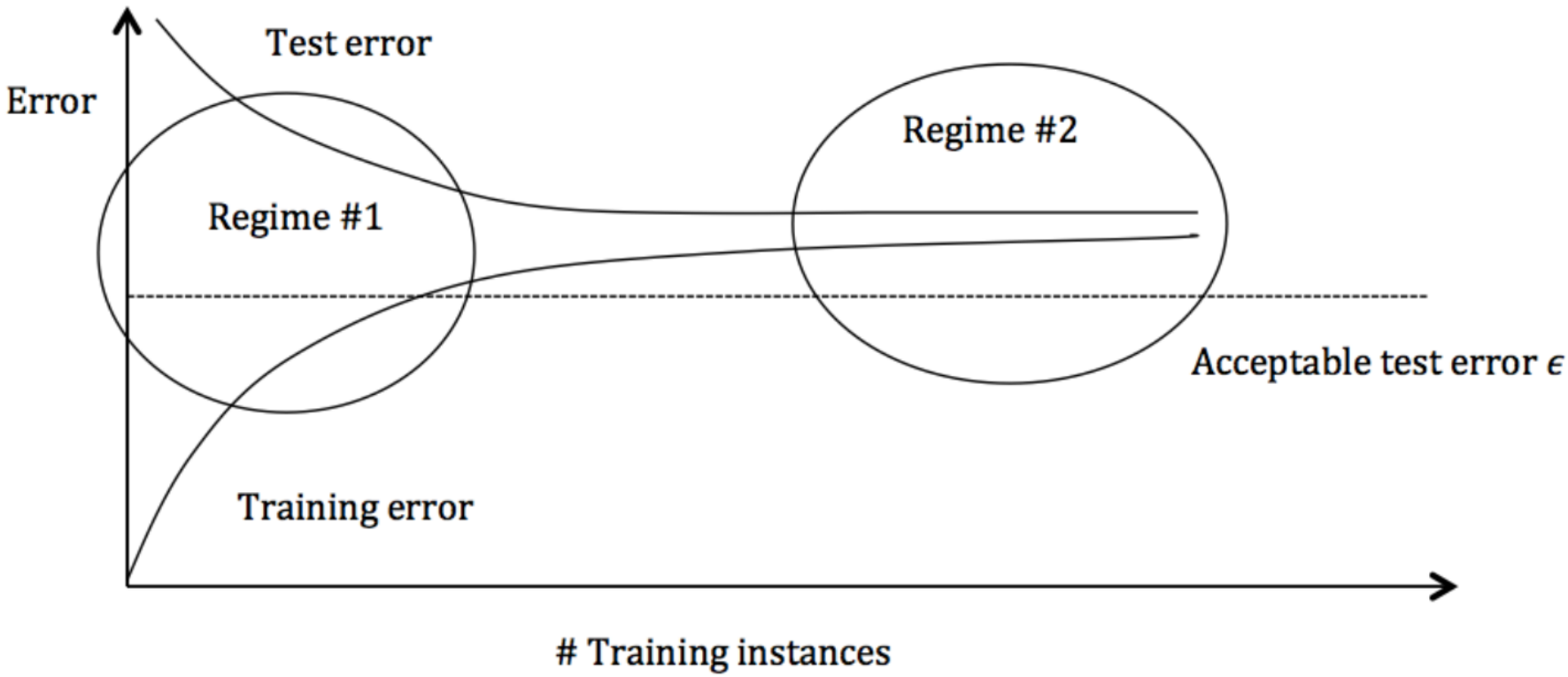


Adapted from Killian Weinberger (Cornell)

Detecting High Variance/Bias



Adapted from Killian Weinberger (Cornell)

Detecting High Variance

Regime 1 (High Variance)

In the first regime, the cause of the poor performance is high variance.

Symptoms:

1. Training error is much lower than test error
2. Training error is lower than ϵ
3. Test error is above ϵ

Remedies:

- Add more training data
- Reduce model complexity -- complex models are prone to high variance
- Bagging (will be covered later in the course)

Adapted from Killian Weinberger (Cornell)

Detecting High Bias

Regime 2 (High Bias)

Unlike the first regime, the second regime indicates high bias: the model being used is not robust enough to produce an accurate prediction.

Symptoms:

1. Training error is higher than ϵ

Remedies:

- Use more complex model (e.g. kernelize, use non-linear models)
- Add features
- Boosting (will be covered later in the course)

Adapted from Killian Weinberger (Cornell)

Reduce Variance Without Increasing Bias

- Averaging reduces variance: let $Z_1, \dots, Z_N$ be i.i.d random variables

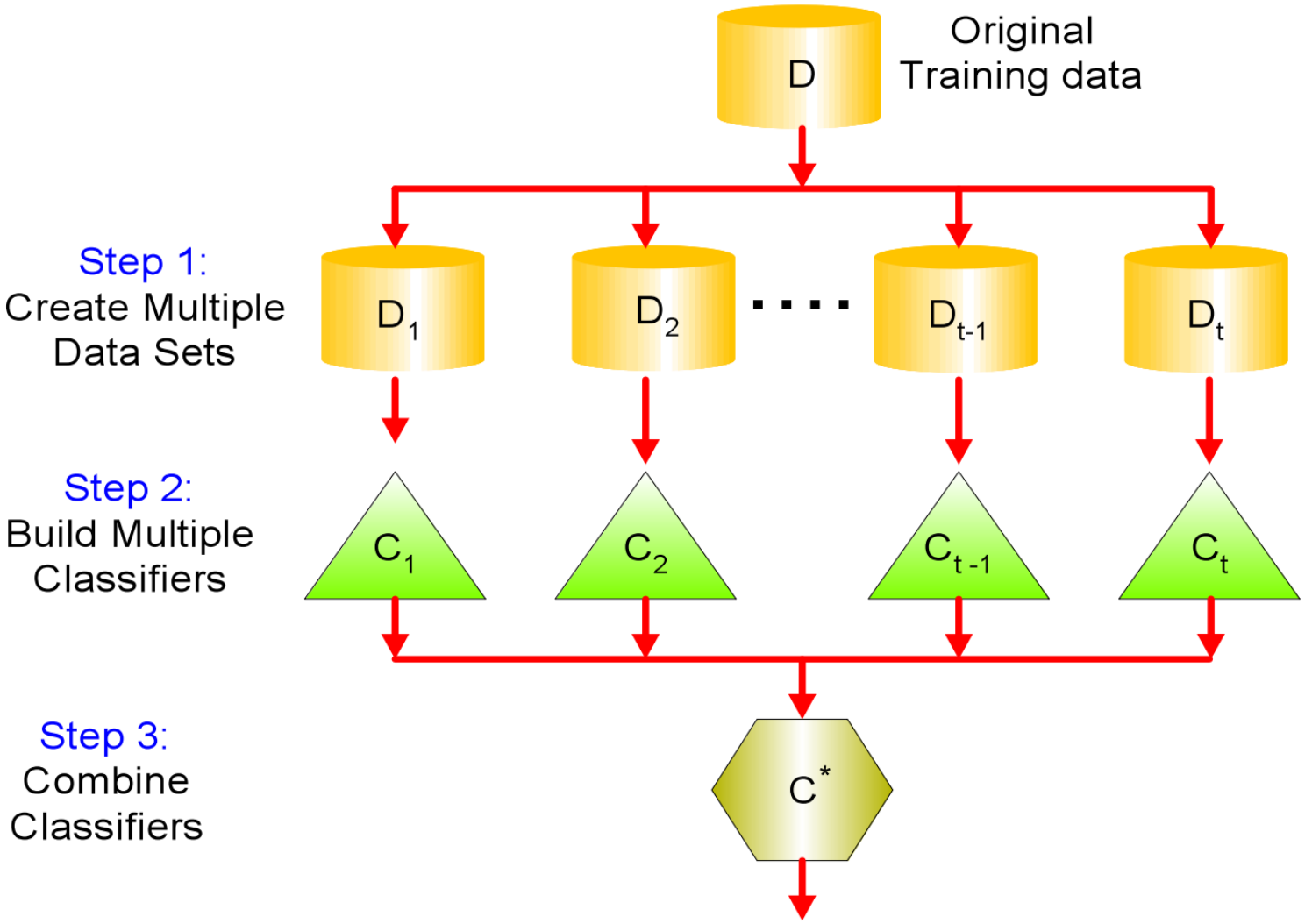
$$Var\left(\frac{1}{N}\sum_i Z_i\right) = \frac{1}{N}Var(Z_i)$$

- Idea: average models to reduce model variance
- How to apply it to our setting?
  - Only one training set
  - Where do multiple models come from?

Bagging: Bootstrap Aggregation

- Take repeated bootstrap samples from training set D (Breiman, 1994)
- Bootstrap sampling: Given set D containing N training examples, create D' by drawing N examples at random **with replacement** from D
- Bagging:
  - Create k bootstrap samples $D_1, \ldots, D_k$
  - Train distinct classifier on each D_i
  - Classify new instance by majority vote / average

Bagging: Bootstrap Aggregation



[image from the slides of David Sontag]

Bagging

Data	1	2	3	4	5	6	7	8	9	10
BS 1	7	1	9	10	7	8	8	4	7	2
BS 2	8	1	3	1	1	9	7	4	10	1
BS 3	5	4	8	8	2	5	5	7	8	8

- Build a classifier from each bootstrap sample
- In each bootstrap sample, each data point has probability $\left(1-\frac{1}{N}\right)^N$ of not being selected
  - Expected number of distinct data points in each sample is then

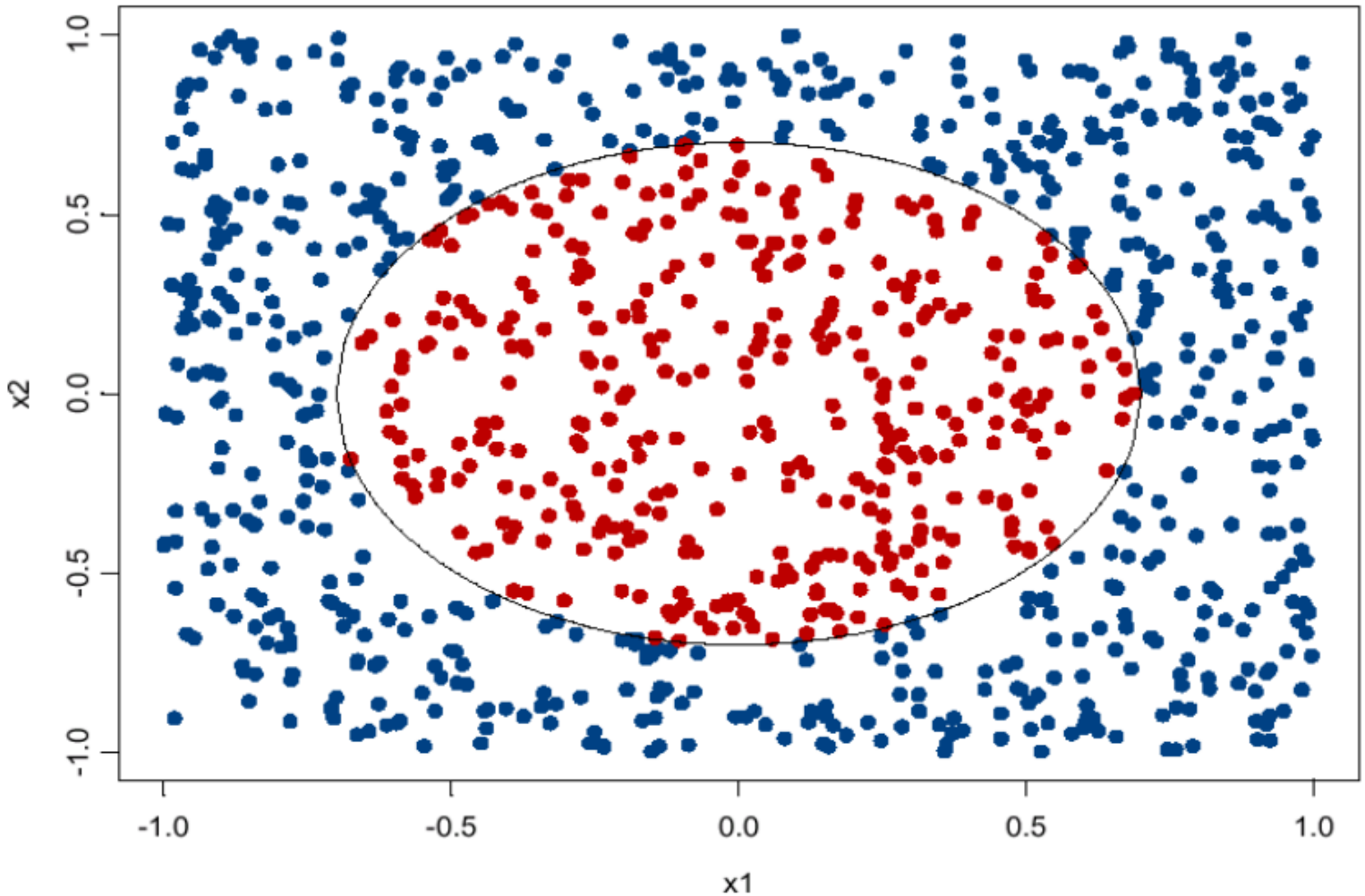
$$N \cdot \left(1-\left(1-\frac{1}{N}\right)^N\right) \approx N \cdot (1-\exp(-1)) = .632 \cdot N$$

Bagging

Data	1	2	3	4	5	6	7	8	9	10
BS 1	7	1	9	10	7	8	8	4	7	2
BS 2	8	1	3	1	1	9	7	4	10	1
BS 3	5	4	8	8	2	5	5	7	8	8

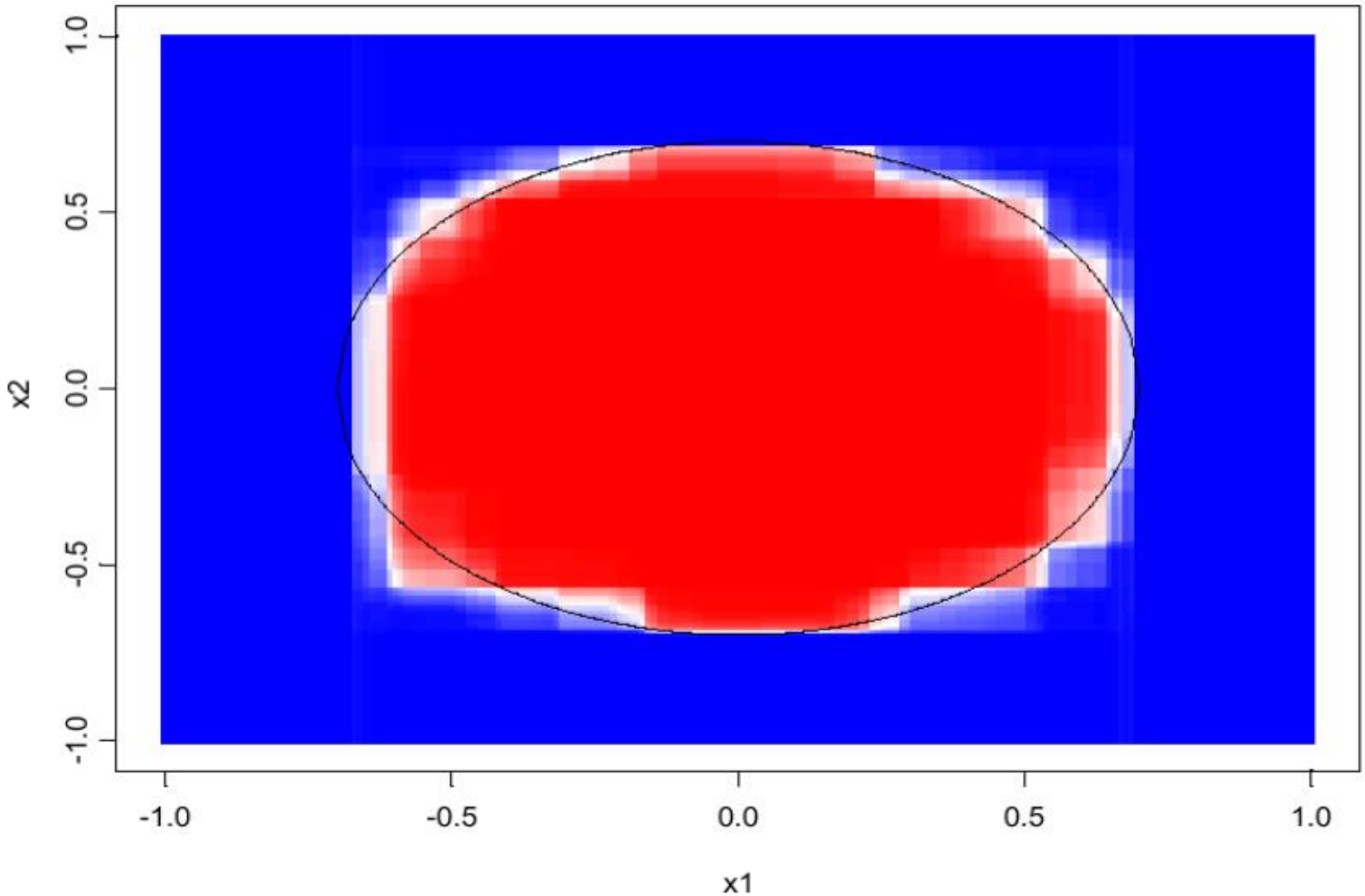
- Build a classifier from each bootstrap sample
- In each bootstrap sample, each data point has probability $\left(1-\frac{1}{N}\right)^N$ of not being selected
  - If we have 1 TB of data, each bootstrap sample will be ~ 632GB (this can present computational challenges, e.g., you shouldn't replicate the data)

Decision Tree Bagging



[image from the slides of David Sontag]

Decision Tree Bagging (100 Bagged Trees)



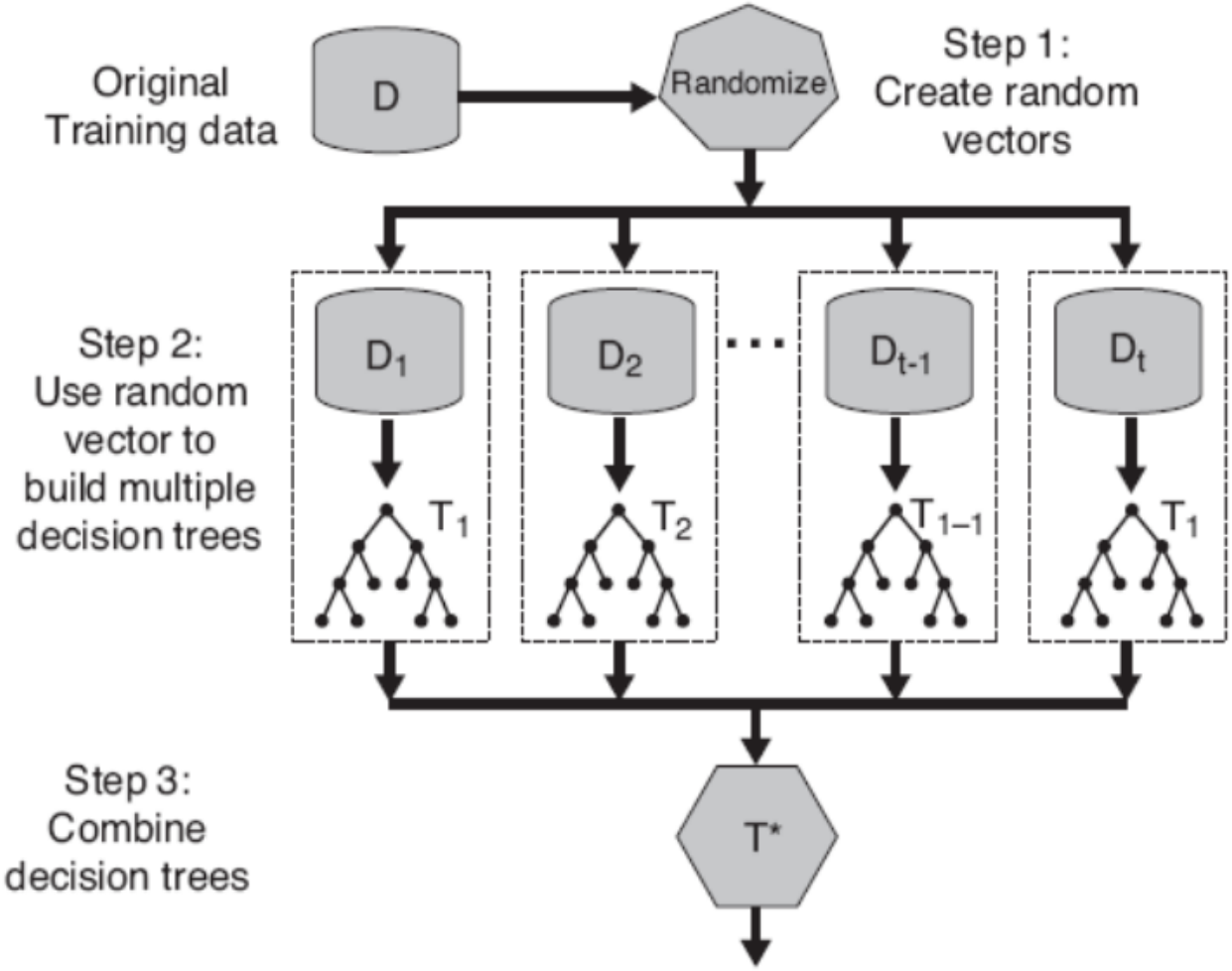
[image from the slides of David Sontag]

Bagging Results

	Without Bagging	With Bagging	
Data Set	$\bar{e}_S$	$\bar{e}_B$	**Decrease**
waveform	29.1	19.3	34%
heart	4.9	2.8	43%
breast cancer	5.9	3.7	37%
ionosphere	11.2	7.9	29%
diabetes	25.3	23.9	6%
glass	30.4	23.6	22%
soybean	8.6	6.8	21%

Breiman "Bagging Predictors" Berkeley Statistics Department TR#421, 1994

Random Forests



Random Forests

- Ensemble method specifically designed for decision tree classifiers
- Introduce two sources of randomness: “bagging” and “random input vectors”
  - Bagging method: each tree is grown using a bootstrap sample of training data
  - Random vector method: best split at each node is chosen from a random sample of m attributes instead of all attributes

Random Forest Algorithm

- For $b = 1$ to B
  - Draw a bootstrap sample of size N from the data
  - Grow a tree T_b using the bootstrap sample as follows
    - Choose m attributes uniformly at random from the data
    - Choose the best attribute among the m to split on
    - Split on the best attribute and recurse (until partitions have fewer than s_{min} number of nodes)

- Prediction for a new data point x
  - Regression: $\frac{1}{B}\sum_b T_b(x)$
  - Classification: choose the majority class label among $T_1(x), \ldots, T_B(x)$

Random Forest Demo

A demo of random forests implemented in JavaScript

When Will Bagging Improve Accuracy?

- Depends on the stability of the base-level classifiers
- A learner is unstable if a small change to the training set causes a large change in the output hypothesis
  - If small changes in D cause large changes in the output, then there will likely be an improvement in performance with bagging
- Bagging can help unstable procedures, but could hurt the performance of stable procedures
  - Decision trees are unstable
  - k-nearest neighbor is stable